```
struct ibv_mr *mr;
 mr = ibv_reg_mr(pd, buf, size, IBV_ACCESS_LOCAL_WRITE);
if (!mr) {
        fprintf(stderr, "Error, ibv_reg_mr() failed\n");
        return -1;
}
```

Creating a completion queue

ibv_create_cq( ) creates a completion queue for an RDMA device context.

An example is given below:

```
struct ibv_cq *cq;
 cq = ibv_create_cq(context, 100, NULL, NULL, 0);
if (!cq) {
        fprintf(stderr, "Error, ibv_create_cq()
failed\n");
        return -1;
}
```

Creating a queue pair

ibv_create_qp( ) creates a queue pair associated with a protection domain.

An example is given below:

```
struct ibv_pd *pd;
struct ibv_cq *cq;
struct ibv_qp *qp;
struct ibv_qp_init_attr qp_init_attr;
memset(&qp_init_attr, 0, sizeof(qp_init_attr));
qp_init_attr.send_cq = cq;
qp_init_attr.recv_cq = cq;
qp_init_attr.qp_type = IBV_QPT_RC;
qp_init_attr.cap.max_send_wr  = 2;
qp_init_attr.cap.max_recv_wr  = 2;
qp_init_attr.cap.max_send_sge = 1;
qp_init_attr.cap.max_recv_sge = 1;
qp = ibv_create_qp(pd, &qp_init_attr);
if (!qp) {
        fprintf(stderr, "Error, ibv_create_qp()
failed\n");
        return -1;
}
```

Creating an address vector

ibv_create_ah( ) creates an address handle associated with a protection domain.

Here is an example of how this is done:

```
struct ibv_pd *pd;
struct ibv_ah *ah;
struct ibv_ah_attr ah_attr;
memset(&ah_attr, 0, sizeof(ah_attr));
ah_attr.is_global     = 0;
ah_attr.dlid          = dlid;
ah_attr.sl            = sl;
ah_attr.src_path_bits = 0;
ah_attr.port_num      = port;
ah = ibv_create_ah(pd, &ah_attr);
if (!ah) {
        fprintf(stderr, "Error, ibv_create_ah()
failed\n");
        return -1;
}
```

Posting work requests

ibv_post_send( ) posts a linked list of work requests to the send queue of a queue pair.

Here is an example:

```
struct ibv_sge sg;
struct ibv_send_wr wr;
struct ibv_send_wr *bad_wr;
memset(&sg, 0, sizeof(sg));
sg.addr          = (uintptr_t)buf_addr;
sg.length = buf_size;
sg.lkey   = mr->lkey;
memset(&wr, 0, sizeof(wr));
wr.wr_id      = 0;
wr.sg_list    = &sg;
wr.num_sge    = 1;
wr.opcode     = IBV_WR_SEND;
wr.send_flags = IBV_SEND_SIGNALED;
if (ibv_post_send(qp, &wr, &bad_wr)) {
        fprintf(stderr, "Error, ibv_post_send()
failed\n");
        return -1;
}
```

Polling for completion

ibv_poll_cq( ) polls work completions from a completion queue.

An example is given below:

```
struct ibv_wc wc;
int num_comp;
do {
        num_comp = ibv_poll_cq(cq, 1, &wc);
} while (num_comp == 0);
if (num_comp < 0) {
        fprintf(stderr, "ibv_poll_cq() failed\n");
        return -1;
}
```

END

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